

BEE 4750 Homework 5: Mixed Integer and Stochastic Programming

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Due Date

Thursday, 12/04/24, 9:00pm

Overview

Instructions

- In Problem 1, you will use mixed integer programming to solve a waste load allocation problem.
- In Problem 2, you will formulate a stochastic optimization problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
Activating project at `~/Desktop/BEE 4750/hw5-mattyb`
Installed PlotUtils          v1.4.4
Installed GR_jll             v0.73.18+0
Installed OpenBLAS32_jll     v0.3.29+0
```

```

Installed Measures          v0.3.3
Installed OpenSSL           v1.6.0
Installed StaticArrays      v1.9.15
Installed MutableArithmetics v1.6.7
Installed HiGHS_jll         v1.12.0+0
Installed Pango_jll         v1.57.0+0
Installed JSON              v1.3.0
Installed StaticArraysCore  v1.4.4
Installed FFMPEG            v0.4.5
Installed Graphs            v1.13.1
Installed DataStructures    v0.19.3
Installed METIS_jll         v5.1.3+0
Installed GraphRecipes      v0.5.15
Installed StatsBase         v0.34.8
Installed ChainRulesCore    v1.26.0
Installed HiGHS             v1.20.1
Installed FFMPEG_jll        v8.0.0+0
Installed SimpleTraits      v0.9.5
Installed Adapt            v4.4.0
Installed Interpolations    v0.16.2
Installed StableRNGs        v1.0.4
Installed ForwardDiff       v1.3.0
Installed JuMP              v1.29.3
Installed StructUtils       v2.6.0
Installed GR                v0.73.18
Precompiling project...
1034.7 ms Measures
842.8 ms StableRNGs
842.0 ms Adapt
841.4 ms StaticArraysCore
567.6 ms SimpleTraits
577.6 ms StructUtils
967.3 ms ChainRulesCore
446.3 ms EarCut_jll
496.7 ms METIS_jll
1944.1 ms DataStructures
558.7 ms OpenBLAS32_jll
831.5 ms ColorVectorSpace → SpecialFunctionsExt
1511.8 ms OpenSSL
1248.7 ms Pango_jll
454.7 ms OffsetArrays → OffsetArraysAdaptExt
704.7 ms Adapt → AdaptSparseArraysExt
1533.1 ms FFMPEG_jll

```

534.5 ms	DiffResults
546.7 ms	StructUtils → StructUtilsTablesExt
4563.0 ms	MutableArithmetics
617.9 ms	ChainRulesCore → ChainRulesCoreSparseArraysExt
3607.4 ms	ColorSchemes
538.8 ms	SortingAlgorithms
805.3 ms	LogExpFunctions → LogExpFunctionsChainRulesCoreExt
1100.8 ms	SpecialFunctions → SpecialFunctionsChainRulesCoreExt
2595.4 ms	JSON
2406.2 ms	HiGHS_jll
4957.0 ms	StaticArrays
2685.7 ms	FFMPEG
3061.3 ms	GR_jll
2865.3 ms	ForwardDiff
2044.0 ms	StatsBase
1309.2 ms	ArnoldiMethod
1472.1 ms	BenchmarkTools
658.5 ms	StaticArrays → StaticArraysChainRulesCoreExt
617.0 ms	StaticArrays → StaticArraysStatisticsExt
687.3 ms	Adapt → AdaptStaticArraysExt
1044.1 ms	ForwardDiff → ForwardDiffStaticArraysExt
8307.3 ms	PlotUtils
1685.3 ms	GeometryTypes
13876.3 ms	HTTP
2280.4 ms	Interpolations
9172.6 ms	GeometryBasics
3572.5 ms	Graphs
2049.5 ms	PlotThemes
2883.7 ms	RecipesPipeline
1108.0 ms	Interpolations → InterpolationsForwardDiffExt
3248.4 ms	GR
1161.6 ms	NetworkLayout
1233.4 ms	NetworkLayout → NetworkLayoutGraphsExt
3487.8 ms	GraphRecipes
34507.2 ms	DataFrames
30127.2 ms	MathOptInterface
1879.2 ms	MathOptIIS
2672.0 ms	Latexify → DataFramesExt
7147.0 ms	HiGHS
12853.3 ms	JuMP
41515.6 ms	Plots
2942.0 ms	Plots → GeometryBasicsExt

59 dependencies successfully precompiled in 72 seconds. 193 already precompiled.

```

using JuMP
using HiGHS
using DataFrames
using GraphRecipes
using Plots
using Measures
using MarkdownTables

```

Problems (Total: 30 Points)

Problem 1 (24 points)

Three cities are developing a coordinated municipal solid waste (MSW) disposal plan. Three disposal alternatives are being considered: a landfill (LF), a materials recycling facility (MRF), and a waste-to-energy facility (WTE). The capacities of these facilities and the fees for operation and disposal are provided below.

- **LF:** Capacity 200 Mg, fixed cost \$2000/day, tipping cost \$50/Mg;
- **MRF:** Capacity 350 Mg, fixed cost \$1500/day, tipping cost \$7/Mg, recycling cost \$40/Mg recycled;
- **WTE:** Capacity 210 Mg, fixed cost \$2500/day, tipping cost \$60/Mg;

Transportation costs are \$1.5/Mg-km, and the relative distances between the cities and facilities are provided in the table below.

City/Facility	Landfill (km)	MRF (km)	WTE (km)
1	5	30	15
2	15	25	10
3	13	45	20
LF	-	32	18
MRF	32	-	15
WTE	18	15	-

The fixed costs associated with the disposal options are incurred only if the particular disposal option is implemented. The three cities produce 100, 90, and 120 Mg/day of solid waste, respectively, with the composition provided in the table below.

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Food Wastes	15	8	0

Component	% of total mass	Combustion ash (%)	MRF Recycling rate (%)
Paper & Cardboard	40	7	55
Plastics	5	5	15
Textiles	3	10	10
Rubber, Leather	2	15	0
Wood	5	2	30
Yard Wastes	18	2	40
Glass	4	100	60
Ferrous	2	100	75
Aluminum	2	100	80
Other Metal	1	100	50
Miscellaneous	3	70	0

The information in the above table will help you determine the overall recycling and ash fractions. Note that the recycling residuals, which may be sent to either landfill or the WTE, have different ash content than the ash content of the original MSW. You will need to determine these fractions to construct your mass balance constraints.

Reminder: Use `round(x; digits=n)` to report values to the appropriate precision!

Problem 1.1

Based on the information above, calculate the overall recycling and ash fractions for the waste produced by each city.

To do this, I store all data from the table into a dictionary, and loop through the dictionary to calculate the weighted averages.

```
waste_data = Dict(
    "Food Wastes" => Dict(
        "percent_mass" => 15.0,
        "combustion_ash" => 8,
        "recycling_rate" => 0
    ),
    "Paper & Cardboard" => Dict(
        "percent_mass" => 40.0,
        "combustion_ash" => 7,
        "recycling_rate" => 55
    ),
    "Plastics" => Dict(
        "percent_mass" => 5.0,
```

```

        "combustion_ash" => 5,
        "recycling_rate" => 15
    ),
    "Textiles" => Dict(
        "percent_mass" => 3.0,
        "combustion_ash" => 10,
        "recycling_rate" => 10
    ),
    "Rubber, Leather" => Dict(
        "percent_mass" => 2.0,
        "combustion_ash" => 15,
        "recycling_rate" => 0
    ),
    "Wood" => Dict(
        "percent_mass" => 5.0,
        "combustion_ash" => 2,
        "recycling_rate" => 30
    ),
    "Yard Wastes" => Dict(
        "percent_mass" => 18.0,
        "combustion_ash" => 2,
        "recycling_rate" => 40
    ),
    "Glass" => Dict(
        "percent_mass" => 4.0,
        "combustion_ash" => 100,
        "recycling_rate" => 60
    ),
    "Ferrous" => Dict(
        "percent_mass" => 2.0,
        "combustion_ash" => 100,
        "recycling_rate" => 75
    ),
    "Aluminum" => Dict(
        "percent_mass" => 2.0,
        "combustion_ash" => 100,
        "recycling_rate" => 80
    ),
    "Other Metal" => Dict(
        "percent_mass" => 1.0,
        "combustion_ash" => 100,
        "recycling_rate" => 50
    )

```

```

    ),
    "Miscellaneous" => Dict(
        "percent_mass" => 3.0,
        "combustion_ash" => 70,
        "recycling_rate" => 0
    )
)

overall_recycling_rate = 0
overall_ash_fraction = 0
for component in values(waste_data)
    overall_recycling_rate += component["percent_mass"]/100 * component["recycling_rate"]
    overall_ash_fraction += component["percent_mass"]/100 * component["combustion_ash"]
end

println("Overall recycling rate: ", overall_recycling_rate, "%")
println("Overall ash fraction: ", overall_ash_fraction, "%")

nonrecycled_waste_data = deepcopy(waste_data)
recycling_residuals_ash_fraction = 0
for component in values(nonrecycled_waste_data)
    component["percent_mass"] = component["percent_mass"] * (1-component["recycling_rate"])/100
    recycling_residuals_ash_fraction += component["percent_mass"]/100 * component["combustion_ash"]
end

println()
println("For later parts:")
println("Recycling residuals ash fraction: ", round(recycling_residuals_ash_fraction; digits=2), "%")

```

Overall recycling rate: 37.75%

Overall ash fraction: 16.41%

For later parts:

Recycling residuals ash fraction: 13.86%

Problem 1.2

What are the decision variables for your optimization problem? Provide notation and variable meaning.

Same as decision variables as the lecture slides:

W_{ij} = Waste transported from city i to disposal j (Mg/day)

R_{kj} = Residual waste transported from disposal k to disposal j (Mg/day)

Y_j = Operational status (on/off) of disposal j (binary)

Problem 1.3

Formulate the objective function. Make sure to include any needed derivations or justifications for your equation(s).

Same objective as the lecture slides:

Minimize total cost

Minimize transportation costs + disposal costs

Let

a_{ij} = Cost of transporting waste from source i to disposal j (\$/Mg-km)

l_{ij} = Distance between source i and disposal j (km)

c_j = Fixed costs of operating disposal j (\$/day)

b_j = Variable cost of disposing waste at disposal j (\$/Mg)

Then

$$\text{transportation costs} = \sum_{i \in I, j \in J} a_{ij} l_{ij} W_{ij}$$

$$\text{disposal costs} = \sum_{j \in J} [c_j Y_j + b_j \sum_i W_{ij}]$$

Objective

$$\min_{W_{ij}, Y_j} \sum_i \sum_j a_{ij} l_{ij} W_{ij} + \sum_j [c_j Y_j + b_j \sum_i W_{ij}]$$

Inputting Values

$$\text{WTE disposal costs} = 2500Y_{WTE} + 60(W_{1,WTE} + W_{2,WTE} + W_{3,WTE} + R_{MRF,WTE})$$

$$\text{MRF disposal costs} = 1500Y_{MRF} + 7(W_{1,MRF} + W_{2,MRF} + W_{3,MRF}) + 0.3775(40)(W_{1,MRF} + W_{2,MRF} + W_{3,MRF})$$

$$\text{LF disposal costs} = 2000Y_{LF} + 50(W_{1,LF} + W_{2,LF} + W_{3,LF} + R_{MRF,LF} + R_{WTE,LF})$$

$$\begin{aligned} \text{Transportation costs} = & 1.5(15W_{1,WTE} + 10W_{2,WTE} + 20W_{3,WTE} + 30W_{1,MRF} + 25W_{2,MRF} \\ & + 45W_{3,MRF} + 5W_{1,LF} + 15W_{2,LF} + 13W_{3,LF} + 15R_{MRF,WTE} + 32R_{MRF,LF} + 18R_{WTE,LF}) \end{aligned}$$

Final objective (calculating total cost and simplifying)

$$\begin{aligned} \min_{W_{ij}, R_{ij}, Y_j} & 2500Y_{WTE} + 1500Y_{MRF} + 2000Y_{LF} + 82.5W_{1,WTE} + 75W_{2,WTE} + 90W_{3,WTE} + 67.1W_{1,MRF} \\ & + 59.6W_{2,MRF} + 89.6W_{3,MRF} + 57.5W_{1,LF} + 72.5W_{2,LF} + 69.5W_{3,LF} + 82.5R_{MRF,WTE} + 98R_{MRF,LF} + 77R_{WTE,LF} \end{aligned}$$

Problem 1.4

Derive all relevant constraints. Make sure to include any needed justifications or derivations.

There are 5 categories of constraints

- 1) Indicator variables are 1 or 0 depending on if the facility is operating

$$Y_j = \begin{cases} 0 & \text{if } \sum_{i \in I} W_{ij} + \sum_{k \neq j} R_{kj} = 0 \\ 1 & \text{if } \sum_{i \in I} W_{ij} + \sum_{k \neq j} R_{kj} > 0 \end{cases}$$

2) All waste from each source needs to be disposed

$$\sum_i W_{ij} + \sum_k R_{kj} \leq K_j$$

3) Non-recycled and residual ash mass balance

$$R_{MRF,WTE} + R_{MRF,LF} = \text{recycling}\% * 5(W_{1,MRF} + W_{2,MRF} + W_{3,MRF})$$

$$R_{WTE,LF} = \text{ash}\% * (W_{1,WTE} + W_{2,WTE} + W_{3,WTE}) + \text{residual ash}\% * (R_{MRF,WTE})$$

4) Disposal facilities can't go above capacity

$$\sum_j W_{ij} = S_i$$

5) Non-negative constraint

$$W_{ij}, R_{ij} \geq 0$$

Inputting values

1.

$$Y_{WTE} = \begin{cases} 0 & \text{if } W_{1,WTE} + W_{2,WTE} + W_{3,WTE} + R_{MRF,WTE} = 0 \\ 1 & \text{else} \end{cases}$$

$$Y_{MRF} = \begin{cases} 0 & \text{if } W_{1,MRF} + W_{2,MRF} + W_{3,MRF} = 0 \\ 1 & \text{else} \end{cases}$$

$$Y_{LF} = 1$$

2.

$$W_{1,WTE} + W_{1,MRF} + W_{1,LF} = 100$$

$$W_{2,WTE} + W_{2,MRF} + W_{2,LF} = 90$$

$$W_{3,WTE} + W_{3,MRF} + W_{3,LF} = 120$$

3.

$$R_{MRF,WTE} + R_{MRF,LF} = 0.6225(W_{1,MRF} + W_{2,MRF} + W_{3,MRF})$$

$$R_{WTE,LF} = 0.1641(W_{1,WTE} + W_{2,WTE} + W_{3,WTE}) + 0.1386(R_{MRF,WTE})$$

4.

$$W_{1,WTE} + W_{2,WTE} + W_{3,WTE} + R_{MRF,WTE} \leq 210$$

$$W_{1,MRF} + W_{2,MRF} + W_{3,MRF} \leq 350$$

$$W_{1,LF} + W_{2,LF} + W_{3,LF} + R_{MRF,LF} + R_{WTE,LF} \leq 200$$

5.

$$W_{ij}, R_{ij} \geq 0$$

Problem 1.5

Find the optimal solution (using JuMP to solve the problem). Report the optimal objective value.

```
using JuMP
using HiGHS

# -----
# Model
# -----
model = Model(HiGHS.Optimizer)
set_silent(model)

# Sets
i = 1:3                # waste types: W1, W2, W3
fac = [:WTE, :MRF, :LF] # facilities

# -----
# Decision Variables
# -----
@variables(model, begin
    W[i, fac] >= 0      # waste flows
    R_MRF_WTE >= 0      # residuals from MRF to WTE
    R_MRF_LF >= 0       # residuals from MRF to LF
    R_WTE_LF >= 0       # residuals from WTE to LF
    Y_WTE, Bin
    Y_MRF, Bin
    Y_LF == 1           # specified in problem
end)

# -----
# Objective
# -----
```

```

@objective(model, Min,
    2500*Y_WTE
  + 1500*Y_MRF
  + 2000*Y_LF
  + 82.5*W[1,:WTE] + 75*W[2,:WTE] + 90*W[3,:WTE]
  + 67.1*W[1,:MRF] + 59.6*W[2,:MRF] + 89.6*W[3,:MRF]
  + 57.5*W[1,:LF] + 72.5*W[2,:LF] + 69.5*W[3,:LF]
  + 82.5*R_MRF_WTE
  + 98*R_MRF_LF
  + 77*R_WTE_LF
)

# -----
# Mass Balance
# -----
@constraint(model, W[1,:WTE] + W[1,:MRF] + W[1,:LF] == 100)
@constraint(model, W[2,:WTE] + W[2,:MRF] + W[2,:LF] == 90)
@constraint(model, W[3,:WTE] + W[3,:MRF] + W[3,:LF] == 120)

# -----
# Corrected Recycling Relations
# -----
@constraint(model,
    R_MRF_WTE + R_MRF_LF ==
    0.6225 * (W[1,:MRF] + W[2,:MRF] + W[3,:MRF])
)

@constraint(model,
    R_WTE_LF ==
    0.1641 * (W[1,:WTE] + W[2,:WTE] + W[3,:WTE]) +
    0.1386 * R_MRF_WTE
)

# -----
# Facility capacity limits
# -----
@constraint(model,
    W[1,:WTE] + W[2,:WTE] + W[3,:WTE] + R_MRF_WTE 210)

@constraint(model,
    W[1,:MRF] + W[2,:MRF] + W[3,:MRF] 350)

```

```

@constraint(model,
    W[1,:LF] + W[2,:LF] + W[3,:LF] + R_MRF_LF + R_WTE_LF 200)

# -----
# Binary Y Logic
# -----
M = 1e5

# WTE activation
@constraint(model,
    sum(W[i,:WTE] for i in 1:3) + R_MRF_WTE M * Y_WTE)
@constraint(model,
    sum(W[i,:WTE] for i in 1:3) + R_MRF_WTE 1e-6 * Y_WTE)

# MRF activation
@constraint(model,
    sum(W[i,:MRF] for i in 1:3) M * Y_MRF)
@constraint(model,
    sum(W[i,:MRF] for i in 1:3) 1e-6 * Y_MRF)

# -----
# Solve
# -----
optimize!(model)

# -----
# Output: ONLY objective + variable values
# -----
println("Objective Value = ", objective_value(model))
for v in all_variables(model)
    println(name(v), " = ", value(v))
end

```

```

Objective Value = 27855.48211508554
W[1,WTE] = 0.0
W[2,WTE] = 90.0
W[3,WTE] = 41.594688359851666
W[1,MRF] = 0.0
W[2,MRF] = -0.0
W[3,MRF] = 0.0
W[1,LF] = 100.0
W[2,LF] = -0.0

```

```

W[3,LF] = 78.40531164014834
R_MRF_WTE = 0.0
R_MRF_LF = 0.0
R_WTE_LF = 21.59468835985166
Y_WTE = 1.0
Y_MRF = -0.0
Y_LF = 1.0

```

This was solved by inputting images of the linear program into ChatGPT, and adjusting the ChatGPT code accordingly.

Optimal total cost: \$27,855

Problem 1.6

Draw a diagram showing the flows of waste between the cities and the facilities. Which facilities (if any) will not be used? Does this solution make sense?

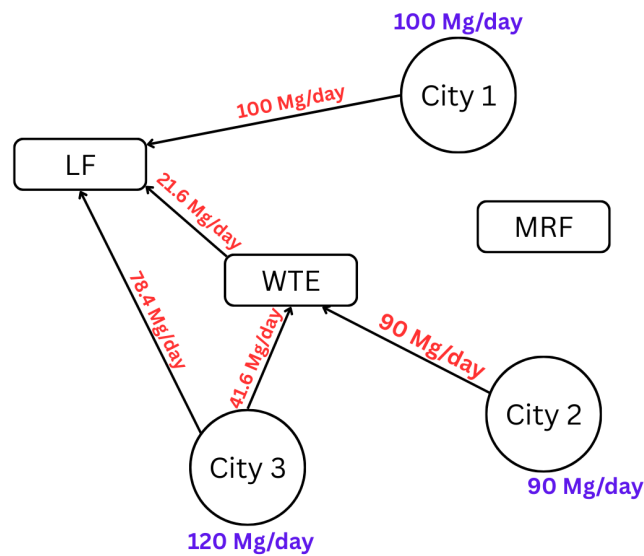


Figure 1: Figure 1: Flows of waste between cities and facilities

Problem 2 (6 points)

Consider a two-period economic dispatch problem, based on the multi-period example from Lecture 14 (on 10/29). The generator data, including ramping constraints for each generator, is provided in ‘data/generators.csv.’ In period 1, the demand is $d_1 = 1100\text{MW}$. In period 2, the demand is projected to be $d_2 = 1200\text{MW}$, but there is a 25% probability that it is \$1500 . In the first period, the solar capacity factor is 0.9 and the wind capacity factor is 0.45, but in the second period, there is some uncertainty: the forecasted solar and wind capacity factors are 0.95 and 0.4, respectively, but there is a 30% probability that they are 0.75 and 0.5. Your goal is to identify how to dispatch your generators to minimize the cost of meeting demand.

Problem 2.1

Draw a scenario tree for this problem.

Problem 2.2

Formulate a stochastic linear program for this problem based on your scenario tree from Problem 2.1 and the data in `data/generators.csv`.

References

List any external references consulted, including classmates.